

Reflow Algorithm

Initialization

Initial Distribution

$$X_0^{(0)} \sim \pi_0$$

Target Distribution

$$X_1^{(0)} \sim \pi_1$$

Reflow Iteration

Linear Interpolation

$$X_t = (1-t)X_0^{(k)} + tX_1^{(k)}$$

$$\frac{dX_t}{dt} = X_1^{(k)} - X_0^{(k)}$$

Velocity Field Learning

$$L_v = E_{t,x} [\|v_\theta(x, t) - \frac{dX_t}{dt}\|^2]$$

Train Network

$$v_\theta(x, t)$$

ODE Solving

$$\frac{dZ_t}{dt} = v_\theta(Z_t, t)$$

1-Rectified Flow:
First flow model
learned from
training data

2-Rectified Flow:
Reflow model trained
using samples from
1-rectified flow
for straighter paths

Distribution Coupling Update

$$X_0^{(k+1)} = Z_0^{(k)}$$

$$X_1^{(k+1)} = Z_1^{(k)}$$

Completed
K iterations?

Yes

Sample Start Point

$$Z_0 \sim \pi_0$$

Apply Trained
Velocity Field

Generate Target Samples

$$Z_1 \sim \pi_1$$

Legend

Data
Distributions

Processing
Steps

Model
Operations

Decision
Points

Notes